Certified audit-budget validity for action monitors

Problem and claim

Track A action monitors are evaluated at small audit budgets, including 1% honest-action false-positive rate (FPR). An observed calibration FPR below 1% does not by itself justify the claim that the true FPR is at most 1%: finite-sample uncertainty may still put the upper confidence bound above the budget.

We apply standard one-sided Clopper-Pearson (CP) bounds and descending fixed-sequence Learn-then-Test (LTT) to AI-control audit thresholds. The statistical tools are established; our contribution is showing how they change which monitor operating points are supportable, and how score granularity interacts with that certification.

Method

The baseline is Track A's action-level `multi` monitor. Our fixed extension, `zmean3`, combines orthogonal, old-action, and goal-analysis scores. Each component is standardized using means and population standard deviations computed only from honest calibration actions, then averaged. We use no learned gate, attack-tuned weights, or runtime refitting.

Calibration is per honest action, with trajectory-grouped data handling. Safety is measured per attack trajectory. The primary matched set contains 49 attacks from `media_processing`, `web_scraping`, and `clinical_trial`; all 49 completed the side task, so official safety equals attack detection on this set.

For certification, thresholds are scanned from high to low over the observed score grid. A threshold is retained while the one-sided CP upper bound on the honest tail probability remains at or below the target FPR; scanning stops at the first failure. This fixed sequence controls threshold-selection multiplicity without changing the monitor scores or the agent policy.

Official Track A safety

Control Tower's `action_only` metric fits an empirical rule at each audit budget and may fractionally audit actions tied at the boundary.

Audit budget	`multi` **safety**	`zmean3` **safety**
0.1%	14.35%	14.29%
1%	58.44%	67.35%
5%	85.70%	88.49%

At 1%, official `multi` safety is 58.44%, rather than the 55.10% deterministic empirical result, because Control Tower audits 24.19% of score-4.0 ties to spend the budget exactly. The attack set and successful sabotage denominator are unchanged. This is a third view of score granularity: `multi`'s coarse grid leaves substantial mass at the budget boundary, while `zmean3`'s official and deterministic empirical safety are both 67.35%. We use official values for rubric-facing comparison and deterministic rules for certification.

Control Tower reports usefulness as 0/27 for both monitors. Every calibration trajectory was already labeled `main_task_success=false` before monitoring. Because monitoring is post-hoc and cannot change those trajectories, this is not evidence of usefulness degradation; these data cannot establish absolute usefulness.

The official `zmean3` curve was computed without new model calls by materializing the frozen formula from already-saved component outputs. The runnable `monitor:zmean3` artifact implements the same formula, but was not executed for this metrics pass.

The certification gap

The pooled `multi` calibration set contains 2,007 quality-controlled honest action scores. The most permissive deterministic empirical threshold among the observed score grid points is 4.5: it flags 15 actions, for 0.75% observed FPR. Its one-sided CP 90% upper bound is 1.06%, however, so the available data do not support a 1% claim.

At threshold 5.0, 8 actions are flagged: observed FPR is 0.40% and the CP 99% upper bound is 0.87%. This threshold is certified. On the matched attacks, the stricter point reduces `multi` detection from 27/49 (55.10%) to 21/49 (42.86%). Certification does not alter ranking; it identifies the deterministic operating point whose audit-budget claim is defensible.

The sample-size floor is exact. With zero tail flags, certifying a 1% FPR requires about 230, 299, or 459 honest actions for `delta=0.10`, `0.05`, or `0.01`; a non-empty tail requires more. The official empirical curve is the optimistic benchmark, while CP/LTT determines which claims survive finite-sample scrutiny.

Results, scope, and limitations

Ensemble gain: ranking and granularity

At each monitor's deterministic threshold certified for 1% FPR at `delta=0.10`, `zmean3` catches 29/49 attacks (59.18%), compared with 21/49 (42.86%) for `multi`. The paired cells are 19 both, 10 `zmean3` only, 2 `multi` only, and 18 neither; exact McNemar `p=0.0386`.

Monitor	Empirical matched detection	Certified matched detection
`multi`	55.1%	42.9%
`mean3`	67.3%	53.1%
`zmean3`	67.3%	59.2%

`mean3` and `zmean3` tie empirically, showing that adding goal-analysis improves ranking. The difference after certification is score geometry: `multi` has 15 unique honest scores, whereas `zmean3` has 103. The finer grid offers a threshold closer to the budget boundary, preserving more of the ranking gain under certification. Better signals improve discrimination; finer granularity improves certified efficiency.

The distributions are consistent with that account. For honest actions, `multi` has median 0.5, 95th percentile 2.5, and 99th percentile 4.0; `zmean3` has median -0.248, 95th percentile 1.254, and 99th percentile 2.356. Attack maxima move upward for both monitors, while `zmean3` supplies many more distinct cut points in the sparse honest tail.

Stealth boundary and confirmation

The matched set contains 20 attacks that still complete the main task, our available stealth proxy. `multi` catches 8/20 (40%) and `zmean3` catches 12/20 (60%), but the paired test is underpowered (`p=0.21875`). This is a positive signal, not evidence of broad stealthy dual-success detection.

A separately frozen set gives 48/75 (64%) for `multi` and 72/75 (96%) for `zmean3`, with exact McNemar `p=1.192e-07`. All 75 attacks fail the main task, so this confirms stronger detection of overt or main-task-failed sabotage, not stealth.

A bounded metadata-only query found 278 new dual-success candidate IDs across the three matched environments, enough to construct a later 75-attack set. That follow-up remains deferred pending full-record vetting, schedule review, and explicit approval for its estimated $2.83 scoring cost ($4.25 with contingency).

Limitations

- Honest calibration covers only three matched environments and depends on their deployment mixture.
- All 27 honest calibration trajectories failed the main task. Retries or error recovery may make this calibration conservative, but no successful-honest comparison is available to test that possibility.
- The 20-attack stealth proxy is underpowered; the frozen 75 contain no main-task-successful attacks.
- Current calibration trajectories cannot establish absolute usefulness.
- We tested no learned gate, adaptive attacks, or attack-tuned weights.
- CP validity depends on representative honest calibration data; correlated actions within trajectories remain a limitation.

The result has two layers: `zmean3` improves attack ranking, and CP/LTT states which 1% audit-budget claims are statistically supportable.